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Quiz 4

SWE4401 SRS

Software Requirements Specification

for MediBook

Version 1.0

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and nonfunctional requirements for **MediBook Version 1.0**, a web-based hospital appointment booking and management system for City General Hospital, Bangladesh. The system supports patient registration, doctor discovery, appointment booking and payment, receptionist-assisted booking, doctor schedule and diagnosis-note management, administrator maintenance of doctor profiles, automated notifications, and patient medical-history access.

MediBook covers the appointment workflow from registration and doctor discovery through slot reservation, payment, appointment confirmation, notification, consultation, and recording of diagnosis notes. Payment processing and transactional email delivery are provided through external services; these services are interfaces and dependencies of MediBook rather than components of the MediBook application itself.

1.2 Document Conventions

Functional requirements are recorded as **user stories**, with acceptance criteria written in **Given/When/Then** form. Priorities use **MoSCoW** labels, and story estimates use the modified Fibonacci scale (1, 2, 3, 5, 8, 13). These conventions are used consistently throughout the backlog so that requirements and their verification can be traced to the same story.

1.3 Intended Audience and Reading Suggestions

This document is intended for hospital management, hospital IT administration, system administrators, doctors, receptionists, developers, QA/testers, project managers, and technical writers. Hospital staff should begin with Sections 1 and 2 to understand the purpose, scope, users, and operating environment. Developers and architects should then review Section 3 for interfaces and Section 4 for the detailed functional backlog. QA/testers should use Section 4 acceptance criteria and Section 7 verification methods to derive test cases. Sections 5 and 6 define the nonfunctional, data, legal, and operational requirements.

1.4 Product Scope

MediBook provides City General Hospital with a centralized web-based appointment workflow. Patients can search for doctors, view availability, reserve appointment slots, make payments, receive confirmations and reminders, and view their own medical history. Doctors can view their upcoming schedules and record diagnosis notes. Receptionists can perform assisted booking for patients who cannot conveniently use the online system, while system administrators maintain doctor profiles and availability.

The system is intended to reduce manual booking effort, improve visibility of appointment availability, reduce double-booking risk, and provide a consistent payment and confirmation process. MediBook does not replace broader hospital information, accounting, or clinical systems that are outside the scope of this SRS.

1.5 References

- Payment Gateway API documentation for the provider selected by City General Hospital (bKash or SSLCommerz) – provider-specific version to be confirmed in TBD-01.
- Transactional Email Service API/SMTP documentation – provider to be selected and documented under TBD-02.

- City General Hospital data privacy and record-retention policy – hospital policy document and retention period to be confirmed under TBD-09.

2 Overall Description

2.1 Product Perspective

MediBook is a new standalone web application for City General Hospital. It manages user accounts, doctor profiles, availability, appointments, payment status, booking notes, diagnosis notes, and patient medical-history views within the MediBook application. It depends on two external service categories for specialized operations:

- **Payment Gateway** (bKash or SSLCommerz) – processes the patient's selected payment method and returns a payment result and transaction reference to MediBook. The gateway is not part of the MediBook application.
- **Email Notification Service** – an external transactional email API or SMTP service used to deliver booking confirmations and appointment reminders. The email provider is not part of the MediBook application.

MediBook is therefore self-contained with respect to its application and business data, but it is intentionally dependent on these external services for payment and email delivery.

2.2 Product Functions

- Patient self-registration and secure login
- Doctor login and profile-linked schedule access
- Receptionist login for assisted booking
- Administrator login for doctor profile management
- Doctor search and filtering by name or specialization
- Appointment slot browsing (30-day availability calendar)
- Real-time slot locking to prevent double-booking
- Online payment processing via external Payment Gateway
- Automatic slot release on payment failure
- Automated email confirmations and reminders
- Doctor schedule viewing (today + next 7 days) and diagnosis note entry
- Patient medical history viewing

2.3 User Classes and Characteristics

User Class	Characteristics
Patient	Primary, high-frequency user. General public, variable technical expertise. Registers, searches doctors, books and pays for appointments, views medical history.
Doctor	Secondary user, moderate frequency (daily). Views appointment schedule and enters diagnosis notes. Read-only with respect to appointment scheduling.
Receptionist	Secondary user, high frequency. Hospital staff who books and manages appointments on behalf of patients without internet access or who find the system difficult to use.
System Administrator	Low-frequency, high-privilege user. Adds new doctors and maintains doctor profiles. Does not interact with patients or appointments.
Payment Gateway (external)	Non-human external service with no MediBook login. Processes payments through a provider such as bKash or SSLCommerz and returns the payment result.
Email Notification Service (external)	Non-human external actor with no login. Sends confirmation and reminder emails.

2.4 Operating Environment

MediBook is a responsive web application accessible through current supported versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari on desktop, laptop, and tablet devices. The application communicates with clients over HTTPS and is intended to run in a cloud-hosted or equivalent server environment. The production hosting provider, server operating system, database technology, and infrastructure configuration are tracked as TBD-07.

2.5 Design and Implementation Constraints

- Online payments must be handled through the selected external Payment Gateway, either bKash or SSLCommerz. MediBook must not store raw credit/debit card numbers; card-data handling remains with the selected gateway.
- The selected payment gateway may expose card, mobile-banking, and bank/internet-banking options. MediBook shall use only payment methods supported by the selected gateway integration.
- Currency is Bangladeshi Taka (BDT); all consultation fees and payment amounts are displayed and processed in Taka.
- Bangladesh Standard Time (BST/UTC+6) shall be used for appointment schedule and slot calculations.
- Email delivery depends on the availability of the external Email Notification Service.
- Doctor availability, specialization, and consultation fee are maintained by the System Administrator, not by doctors themselves.
- Passwords must not be stored as plaintext and must use an approved adaptive password-hashing mechanism such as Argon2id or bcrypt.

2.6 User Documentation

The following documentation will be delivered with the product: an in-app FAQ/help page for patients, a receptionist quick-reference guide for assisted booking, a doctor user guide covering schedules and diagnosis notes, and an administrator manual for managing doctor profiles. The final delivery format and whether PDF copies are required are tracked as TBD-13.

2.7 Assumptions and Dependencies

- Patients have access to a valid email address and mobile number at the time of registration.
- The selected Payment Gateway API is available and returns a definitive transaction result through its supported response/callback mechanism.
- The Email Notification Service is available for outbound confirmation and reminder emails; temporary provider failures are handled without changing appointment status.
- Doctors' availability, specialization, and consultation fee are accurate and kept current by the Administrator.
- The hospital will provide the credentials, sandbox access, and production configuration required for the selected payment and email services.

3 External Interface Requirements

3.1 User Interfaces

MediBook provides distinct dashboards for Patient, Doctor, Receptionist, and Administrator roles. Common UI elements include consistent navigation, clear action labels, standardized error messages, and responsive layouts. Doctor cards shall show name, specialization, available days, and consultation fee in BDT. The booking interface shall display a 30-day availability calendar with clearly distinguished Available and Unavailable states. Example messages include "This email is already registered.", "Incorrect email or password.", "This appointment slot is no longer available. Please select another slot.", and "You do not have permission to access this page." Detailed visual mockups are outside this SRS and may be maintained in a separate UI specification.

3.2 Hardware Interfaces

No specialized hardware is required. MediBook is accessed from standard client devices (desktop, laptop, or tablet) with an internet connection; no direct hardware-level interfacing is performed by the software.

3.3 Software Interfaces

Interface	Description
Payment Gateway API	External payment interface provided by bKash or SSLCommerz. MediBook sends the transaction information required by the selected gateway and receives the payment result and transaction reference through the provider's documented response/callback mechanism.
Email Notification Service	External transactional email API or SMTP service. MediBook sends booking confirmations and appointment reminders addressed to patients and doctors.
Application Database	Persistent database storing user accounts, roles, doctor profiles, availability, appointments, payment status, booking notes, diagnosis notes, and relevant notification records. The production database technology is tracked under TBD-07.

3.4 Communications Interfaces

All client-server communication shall use HTTPS with a supported TLS configuration. Outbound email shall use SMTP over TLS or the selected transactional email provider's HTTPS API. Payment Gateway communication shall use HTTPS and follow the request, response, and callback/webhook rules defined by the selected provider. Payment and email messages shall contain only the data required by the relevant external service. The exact endpoint and message format will be finalized after the provider selections in TBD-01 and TBD-02.

4 Epics and User Stories

4.1 Story Map / Backlog Overview

Epic	User Stories
EPIC-1: Account Registration & Authentication	US-1.1 Patient Registration, US-1.2 User Login, US-1.3 Role-Based Access
EPIC-2: Doctor Search & Discovery	US-2.1 Browse Doctor List, US-2.2 Search/Filter Doctors, US-2.3 View Doctor Availability
EPIC-3: Appointment Booking	US-3.1 View Availability Calendar, US-3.2 Reserve Appointment Slot, US-3.3 Add Booking Note
EPIC-4: Payment Processing	US-4.1 Initiate Payment, US-4.2 Confirm Successful Payment, US-4.3 Handle Failed Payment
EPIC-5: Doctor Schedule & Diagnosis Notes	US-5.1 View Upcoming Schedule, US-5.2 View Booking Note, US-5.3 Enter Diagnosis Notes
EPIC-6: Receptionist-Assisted Booking	US-6.1 Receptionist Login, US-6.2 Book Appointment on Behalf of Patient
EPIC-7: Doctor Profile Management (Admin)	US-7.1 Add New Doctor, US-7.2 Update Doctor Profile
EPIC-8: Notifications	US-8.1 Booking Confirmation Email, US-8.2 Appointment Reminder Email
EPIC-9: Patient Medical History	US-9.1 View Medical History, US-9.2 Prevent Cross-Patient Access

Personas / Roles:

Persona/Role	Description
Patient	Any person wishing to consult a doctor; registers, searches, books, pays, and views history.
Doctor	Hospital physician who views their schedule and records diagnosis notes.
Receptionist	Hospital staff who books/manages appointments for patients unable to use the system directly.
Administrator	Hospital IT/operations staff who add and maintain doctor profiles.
Payment Gateway	External, login-less system processing payments (e.g., bKash or SSLCommerz).
Email Notification Service	External, login-less system sending confirmation and reminder emails.

4.2 Epic 1: Account Registration & Authentication**4.2.1 Epic Description and Priority**

Enables new patients to create an account and allows authorized users to securely log in and access only the dashboard and functions permitted for their role. **Priority: Must.** Target release: Sprint 1.

4.2.2 User Stories

US-1.1:	
ID	US-1.1
Epic	EPIC-1
Story	As a new patient, I want to register with my personal details, so that I can access MediBook's booking features.
Priority	Must
Story Points	3
Dependencies	None
Definition of Done	Code reviewed, unit tested, email-uniqueness validation tested, acceptance criteria met, documented

US-1.2:	
ID	US-1.2
Epic	EPIC-1
Story	As an authorized MediBook user, I want to log in with my email and password, so that I can access the dashboard permitted for my role.
Priority	Must
Story Points	2
Dependencies	US-1.1
Definition of Done	Code reviewed, unit tested, role-specific routing tested, acceptance criteria met, documented

US-1.3:	
ID	US-1.3
Epic	EPIC-1
Story	As a hospital administrator, I want role-based access controls to restrict protected functions, so that users cannot access information or operations outside their responsibilities.
Priority	Must
Story Points	5
Dependencies	US-1.2
Definition of Done	Role permissions implemented, unauthorized-access tests passed, code reviewed, acceptance criteria met, documented

4.2.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-1.1	Given a new patient visits MediBook for the first time, When they submit full name, date of birth, gender, mobile number, a unique email, and password, Then a patient account is created and the patient can proceed to log in.
US-1.1	Given a visitor on the Registration page, When they submit an email that is already registered, Then an error message is displayed: "This email is already registered."
US-1.1	Given one or more required registration fields are missing, When the form is submitted, Then MediBook identifies the missing fields and does not create the account.
US-1.2	Given an authorized MediBook user on the Login page, When they enter matching email and password, Then they are redirected to the dashboard appropriate to their role (Patient, Doctor, Receptionist, or Administrator).
US-1.2	Given a registered user on the Login page, When they enter an incorrect email or password, Then an error message "Incorrect email or password." is displayed and they remain on the Login page.
US-1.3	Given a patient attempts to access an administrator-only page, When the request is made, Then MediBook denies access and displays "You do not have permission to access this page."
US-1.3	Given a doctor logs in successfully, When the dashboard loads, Then doctor-specific functionality is displayed and administrator-only functions are unavailable.

4.3 Epic 2: Doctor Search & Discovery

4.3.1 Epic Description and Priority

Allows patients to browse doctors, search by doctor name or specialization, and view availability before booking an appointment. **Priority: Must.** Target release: Sprint 1.

4.3.2 User Stories

US-2.1:	
ID	US-2.1
Epic	EPIC-2
Story	As a patient, I want to browse a list of available doctors, so that I can decide who to consult.
Priority	Must
Story Points	2
Dependencies	US-1.2, US-7.1
Definition of Done	Code reviewed, unit tested, acceptance criteria met, documented

US-2.2:	
ID	US-2.2
Epic	EPIC-2
Story	As a patient, I want to search doctors by name or specialization, so that I can quickly find a relevant doctor.
Priority	Must
Story Points	3
Dependencies	US-2.1
Definition of Done	Code reviewed, unit tested, matching and no-results cases tested, acceptance criteria met, documented

US-2.3:	
ID	US-2.3
Epic	EPIC-2
Story	As a patient, I want to view a doctor's availability for the next 30 days, so that I can choose a suitable appointment date and time.
Priority	Must
Story Points	5
Dependencies	US-2.1, US-7.2
Definition of Done	Code reviewed, unit tested, 30-day calendar tested, timezone behavior verified, acceptance criteria met, documented

4.3.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-2.1	Given doctors have active profiles, When a patient opens the doctor list, Then each doctor's card shows name, specialization, available days, and consultation fee in BDT.
US-2.2	Given a patient is on the doctor list page, When they search by a doctor's name or specialization keyword, Then the list displays matching doctors.
US-2.2	Given no doctor matches the search term, When the search executes, Then MediBook displays a clear no-results message.
US-2.3	Given a doctor has configured availability, When a patient opens the doctor's booking calendar, Then MediBook displays available and unavailable slots for the next 30 days using Bangladesh Standard Time.
US-2.3	Given a slot has already been booked or reserved, When another patient views the calendar, Then that slot is shown as unavailable.

4.4 Epic 3: Appointment Booking

4.4.1 Epic Description and Priority

Provides the core appointment-selection workflow, including availability display, slot reservation, prevention of double booking, and capture of a brief patient booking note. **Priority: Must.**
Target release: Sprint 2.

4.4.2 User Stories

US-3.1:	
ID	US-3.1
Epic	EPIC-3
Story	As a patient, I want to select an available appointment slot, so that I can request a consultation at a suitable date and time.
Priority	Must
Story Points	3
Dependencies	US-2.3
Definition of Done	Code reviewed, unit tested, unavailable-slot handling tested, acceptance criteria met, documented

US-3.2:	
ID	US-3.2
Epic	EPIC-3
Story	As a patient, I want my selected slot to be reserved while I complete payment, so that another patient cannot successfully book the same slot.
Priority	Must
Story Points	8
Dependencies	US-3.1, US-4.1
Definition of Done	Transactional slot reservation implemented, concurrent-booking tests passed, acceptance criteria met, documented

US-3.3:	
ID	US-3.3
Epic	EPIC-3
Story	As a patient, I want to add a brief note when booking, so that the assigned doctor has basic context before the consultation.
Priority	Must
Story Points	3
Dependencies	US-3.1
Definition of Done	Booking-note field implemented, persisted, displayed to assigned doctor, validation tested, acceptance criteria met, documented

4.4.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-3.1	Given an appointment slot is available, When a patient selects it, Then MediBook proceeds with the booking workflow.
US-3.1	Given a slot has become unavailable, When a patient attempts to select it, Then MediBook prevents the booking and displays “This appointment slot is no longer available. Please select another slot.”
US-3.2	Given a patient selects an available slot, When the booking request is processed, Then MediBook reserves the slot, creates an appointment with status Pending Payment, and prevents another booking from successfully claiming that slot.
US-3.2	Given two patients attempt to book the same slot at nearly the same time, When both requests are processed, Then only one request successfully reserves the slot and the other receives an unavailable-slot response.
US-3.3	Given a patient enters a brief booking note, When the appointment is created, Then the note is stored with the appointment and can be displayed to the assigned doctor.

4.5 Epic 4: Payment Processing

4.5.1 Epic Description and Priority

Connects MediBook to the selected external Payment Gateway so that a pending appointment is confirmed only after a successful payment result is received. The production provider shall be bKash or SSLCommerz. **Priority: Must.** Target release: Sprint 2.

4.5.2 User Stories

US-4.1:	
ID	US-4.1
Epic	EPIC-4
Story	As a patient, I want to pay for my appointment using a payment method supported by the selected gateway, so that I can confirm my booking.
Priority	Must
Story Points	5
Dependencies	US-3.2
Definition of Done	Payment-gateway integration implemented, sandbox transaction tested, payment methods documented, acceptance criteria met

US-4.2:	
ID	US-4.2
Epic	EPIC-4
Story	As a patient, I want my appointment to be confirmed after successful payment, so that I know the booking is secured.
Priority	Must
Story Points	5
Dependencies	US-4.1, EPIC-8
Definition of Done	Successful gateway response handled, appointment status updated, digital receipt generated, notification trigger tested

US-4.3:	
ID	US-4.3
Epic	EPIC-4
Story	As a patient, I want a failed payment to release my reserved slot, so that I can retry or select another appointment.
Priority	Must
Story Points	5
Dependencies	US-4.1
Definition of Done	Failed-payment result handled, slot released, error message displayed, retry workflow tested

4.5.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-4.1	Given a booking has status Pending Payment, When the patient chooses a payment method such as card, mobile banking, or bank/internet banking supported by the selected gateway, Then MediBook sends the required transaction information to the Payment Gateway.
US-4.1	Given the selected gateway supports multiple payment methods, When the patient reaches the gateway payment interface, Then the gateway presents the payment methods available under its configured service.
US-4.2	Given the Payment Gateway returns a successful payment result, When MediBook processes the result, Then the appointment status changes to Confirmed and a digital receipt is generated.
US-4.3	Given the Payment Gateway returns a failed payment result, When MediBook processes the result, Then the appointment is not confirmed, the reserved slot is released, and the patient is shown “Payment could not be completed. Your appointment slot has been released. Please try again.”

4.6 Epic 5: Doctor Schedule & Diagnosis Notes

4.6.1 Epic Description and Priority

Gives doctors a read-only view of their appointment schedule and allows them to record diagnosis notes for relevant consultations. Doctors cannot create, modify, or remove appointments through the doctor dashboard. **Priority: Must.** Target release: Sprint 3.

4.6.2 User Stories

US-5.1:	
ID	US-5.1
Epic	EPIC-5
Story	As a doctor, I want to see my appointment schedule for today and the next 7 days, so that I can prepare for my consultations.
Priority	Must
Story Points	3
Dependencies	US-1.2, US-3.2
Definition of Done	Code reviewed, unit tested, correct-doctor filtering tested, read-only scheduling behavior verified

US-5.2:	
ID	US-5.2
Epic	EPIC-5
Story	As a doctor, I want to view the patient's brief booking note, so that I have basic context before the consultation.
Priority	Must
Story Points	2
Dependencies	US-3.3, US-5.1
Definition of Done	Access to the correct doctor's appointments tested, acceptance criteria met, documented

US-5.3:	
ID	US-5.3
Epic	EPIC-5
Story	As a doctor, I want to enter diagnosis notes for a patient's appointment, so that the consultation outcome is recorded in the system.
Priority	Must
Story Points	5
Dependencies	US-5.1
Definition of Done	Diagnosis-note creation and persistence implemented, authorization tested, acceptance criteria met, documented

4.6.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-5.1	Given a doctor has confirmed appointments, When the doctor opens the dashboard, Then MediBook displays today's and the next seven days' appointments with patient name and appointment time.
US-5.1	Given a doctor is viewing an appointment, When the doctor attempts to modify or delete the appointment, Then MediBook prevents the action because scheduling is read-only for doctors.
US-5.2	Given a patient has entered a booking note, When the assigned doctor views that appointment, Then the doctor can read the note.
US-5.3	Given a doctor is authorized to access an assigned appointment, When the doctor submits diagnosis notes, Then MediBook stores the notes against the appointment for later retrieval in the patient's medical history.
US-5.3	Given a doctor attempts to access diagnosis information for an appointment outside the doctor's authorization, When the request is made, Then MediBook denies access.

4.7 Epic 6: Receptionist-Assisted Booking

4.7.1 Epic Description and Priority

Enables authorized hospital receptionists to book appointments on behalf of patients who lack internet access or find the system difficult to use. Assisted bookings follow the same doctor-availability, slot-locking, and payment rules as self-service bookings. **Priority: Should.** Target release: Sprint 3.

4.7.2 User Stories

US-6.1:	
ID	US-6.1
Epic	EPIC-6
Story	As a receptionist, I want to log in securely, so that I can perform authorized assisted-booking activities.
Priority	Should
Story Points	2
Dependencies	US-1.2, US-1.3
Definition of Done	Receptionist authentication and permissions implemented, tests passed, acceptance criteria met

US-6.2:	
ID	US-6.2
Epic	EPIC-6
Story	As a receptionist, I want to book an appointment on behalf of a patient, so that patients who need assistance can still access hospital appointments.
Priority	Should
Story Points	5
Dependencies	US-6.1, EPIC-2, EPIC-3, EPIC-4
Definition of Done	Assisted-booking workflow implemented, slot-locking and payment rules reused, authorization tested, acceptance criteria met

4.7.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-6.1	Given a receptionist has valid credentials, When the receptionist logs in, Then MediBook displays the receptionist dashboard and authorized booking functions.
US-6.2	Given a patient requires assistance, When an authorized receptionist selects an available doctor and slot for that patient, Then MediBook creates the appointment using the same availability and slot-locking rules as a self-service booking.
US-6.2	Given the assisted appointment requires payment, When booking is submitted, Then the appointment follows the configured payment workflow before becoming Confirmed.

4.8 Epic 7: Doctor Profile Management (Admin)

4.8.1 Epic Description and Priority

Allows the System Administrator to add doctors and maintain doctor profiles, including specialization, consultation fee, and availability, so that patient-facing information remains accurate. The administrator does not directly manage patient appointments through this epic. **Priority: Must.** Target release: Sprint 1.

4.8.2 User Stories

US-7.1:	
ID	US-7.1
Epic	EPIC-7
Story	As a system administrator, I want to add a new doctor's profile, so that patients can find and book that doctor.
Priority	Must
Story Points	5
Dependencies	US-1.2, US-1.3
Definition of Done	Doctor creation form implemented, validation tested, profile stored, authorization verified

US-7.2:	
ID	US-7.2
Epic	EPIC-7
Story	As a system administrator, I want to update a doctor's specialization, consultation fee, and availability, so that patients see current information.
Priority	Must
Story Points	5
Dependencies	US-7.1
Definition of Done	Profile editing implemented, updates persisted, unauthorized access blocked, acceptance criteria met

4.8.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-7.1	Given an authenticated system administrator enters valid doctor information, When the administrator submits the form, Then MediBook creates the doctor profile.
US-7.1	Given a non-administrator user attempts to create a doctor profile, When the request is made, Then MediBook denies the operation.
US-7.2	Given a doctor profile exists, When an administrator updates its specialization or consultation fee and saves, Then the updated information is displayed in patient-facing doctor listings.
US-7.2	Given an administrator updates doctor availability, When a patient views the doctor calendar, Then the calendar reflects the updated availability.

4.9 Epic 8: Notifications

4.9.1 Epic Description and Priority

Keeps patients and doctors informed through automated transactional email using the external Email Notification Service. **Priority: Must.** Target release: Sprint 2.

4.9.2 User Stories

US-8.1:	
ID	US-8.1
Epic	EPIC-8
Story	As a patient, I want to receive a booking confirmation email with my appointment details and digital receipt, so that I have a record of my confirmed appointment.
Priority	Must
Story Points	3
Dependencies	US-4.2
Definition of Done	Email integration implemented, confirmation template created, delivery request tested, acceptance criteria met

US-8.2:	
ID	US-8.2
Epic	EPIC-8
Story	As a patient or doctor, I want to receive a reminder email before an upcoming appointment, so that I do not miss the consultation.
Priority	Must
Story Points	5
Dependencies	US-8.1
Definition of Done	Reminder scheduling implemented, patient/doctor recipients verified, failure handling tested, acceptance criteria met

4.9.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-8.1	Given an appointment payment is confirmed, When the confirmation is processed, Then MediBook sends a booking-confirmation email containing appointment details and a digital receipt to the patient through the configured Email Notification Service.
US-8.1	Given the email service accepts the notification request, When delivery processing completes, Then MediBook records the notification request as successfully submitted.
US-8.2	Given an appointment is Confirmed and its scheduled time is approaching, When the configured reminder interval is reached, Then a reminder email is sent to the patient and doctor.
US-8.2	Given the email service is temporarily unavailable, When a reminder cannot be submitted, Then MediBook records the failure without changing the appointment's Confirmed status.

4.10 Epic 9: Patient Medical History

4.10.1 Epic Description and Priority

Allows patients to view a consolidated history of their own past appointments and associated diagnosis notes while enforcing patient-level privacy. **Priority: Should.** Target release: Sprint 4.

4.10.2 User Stories

US-9.1:	
ID	US-9.1
Epic	EPIC-9
Story	As a patient, I want to view my medical history, so that I can review past consultations and diagnoses.
Priority	Should
Story Points	5
Dependencies	US-5.3
Definition of Done	Medical-history page implemented, appointment and diagnosis data displayed, acceptance criteria met, documented

US-9.2:	
ID	US-9.2
Epic	EPIC-9
Story	As a patient, I want my medical history to be visible only to authorized users, so that my medical information remains private.
Priority	Should
Story Points	5
Dependencies	US-9.1, US-1.3
Definition of Done	Ownership filtering and authorization implemented, cross-patient access tests passed, security review completed

4.10.3 Acceptance Criteria

Story ID	Acceptance Criteria
US-9.1	Given a logged-in patient has previous appointments, When they open the Medical History page, Then they see their own past appointments with doctor name, date, and available diagnosis notes.
US-9.1	Given a patient has no past appointments, When they open the Medical History page, Then an appropriate empty-state message is shown.
US-9.2	Given Patient A is logged in, When Patient A attempts to access Patient B's medical history, Then MediBook denies access.
US-9.2	Given an unauthorized user requests diagnosis information, When MediBook evaluates the request, Then the system denies access according to role permissions.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- Doctor search and filter results must be returned within 2 seconds under normal load for at least 95% of requests.
- Doctor availability for the requested 30-day period must be displayed within 2 seconds under normal load for at least 95% of requests.
- When a patient selects an available slot, the reservation/locking operation must complete within 1 second under normal load.
- The slot reservation operation shall use a database transaction or equivalent concurrency-control mechanism so that two simultaneous requests cannot both successfully reserve the

same slot.

- The payment transaction shall remain in Pending Payment until a definitive gateway result is received or the configured payment timeout expires. The timeout duration is tracked as TBD-03.

5.2 Safety Requirements

MediBook is not a diagnostic or life-support system, but incorrect appointment or patient-record information could indirectly affect patient care. The system shall therefore prevent double booking, preserve the association between appointments and patients, avoid changing a confirmed appointment to an unconfirmed state because of a later email-service failure, and ensure diagnosis notes are stored against the correct appointment. Payment confirmation processing shall be designed so that repeated gateway callbacks do not create duplicate confirmations or inconsistent payment states.

5.3 Security Requirements

- All client-server traffic must use HTTPS/TLS.
- Passwords must be stored only as salted, adaptive one-way password hashes using an approved algorithm such as Argon2id or bcrypt. Plain-text passwords must never be stored.
- MediBook must never store raw credit/debit card numbers or CVV values. Payment-card collection and processing shall be delegated to the selected Payment Gateway, such as bKash or SSLCommerz.
- Role-based access control must enforce that Doctors cannot modify appointment schedules, Receptionists cannot alter doctor profiles, and only System Administrators can manage doctor profiles.
- Patient medical history and diagnosis notes must be accessible only to authorized users according to their role. A patient may view their own history; a doctor may access diagnosis information only for appointments the doctor is authorized to handle.
- Authentication and authorization failures shall not disclose unnecessary account or medical information.
- User sessions shall expire after the inactivity period approved by the hospital; the exact duration is tracked as TBD-04.
- Payment integration shall be designed to minimize MediBook's exposure to payment-card data. The exact compliance boundary and responsibilities between MediBook and the selected gateway shall be confirmed under TBD-05.

5.4 Software Quality Attributes

- **Usability:** The responsive interface shall be usable on desktop and tablet devices and shall provide clear, task-oriented booking steps suitable for patients with varied technical experience.
- **Reliability:** Appointment and payment state transitions shall be consistent. Temporary failure of the Email Notification Service shall not cancel or unconfirm a successfully paid appointment.
- **Availability:** The target production availability is 99.5% uptime, subject to formal hospital approval under TBD-06.

- **Maintainability:** Doctor specialization, availability, and consultation-fee information shall be editable by authorized administrators without source-code changes.
- **Interoperability:** Payment and email integrations shall be isolated behind defined interfaces so that the selected provider can be replaced without redesigning the core appointment workflow.
- **Testability:** Registration, authentication, search, slot reservation, concurrent booking, payment success/failure, authorization, diagnosis notes, notifications, and medical-history access shall be covered by repeatable tests.

5.5 Business Rules

- Only System Administrators may add or edit doctor profiles.
- Doctors have read-only access to appointment scheduling and may only add diagnosis notes to authorized consultations.
- Receptionists may book appointments on behalf of patients but may not modify doctor profiles.
- Patients may view their own medical history; the system must not expose one patient's medical history to another patient.
- A slot becomes unavailable when a booking request successfully reserves it and the associated appointment enters Pending Payment.
- A Pending Payment slot is released when the associated payment fails or the configured payment timeout expires.
- A successful payment changes the appointment status to Confirmed.
- Appointment fees are charged in BDT at the consultation-fee rate configured on the doctor's profile at the time the appointment is created.
- Appointment times are calculated and displayed using Bangladesh Standard Time (UTC+6).
- The production payment gateway shall be either bKash or SSLCommerz unless the hospital formally approves a different provider through a requirements change.

6 Other Requirements

6.1 Data Requirements

MediBook shall maintain, at minimum, logical records for users, patients, doctors, doctor availability, appointments, payments, diagnosis notes, booking notes, and notification requests/statuses. Patient and doctor records shall be linked to the appropriate user account and role.

6.2 Appointment Statuses

MediBook shall support the following logical appointment states:

Status	Meaning
Pending Payment	Slot has been reserved and payment has not yet been successfully completed.
Confirmed	Payment has been successfully confirmed and the appointment is secured.
Released	The pending booking did not complete successfully and the slot is available again.
Completed	The consultation has taken place and may have associated diagnosis notes.

6.3 Internationalization

Version 1.0 shall use English as the primary interface language, Bangladeshi Taka (BDT) as the currency, and Bangladesh Standard Time (UTC+6) as the operational timezone. Bangla-language support is tracked as TBD-08.

6.4 Data Retention

Patient accounts, appointments, payment references, booking notes, and diagnosis notes shall be retained according to City General Hospital's approved record-retention policy. The exact retention period is TBD-09.

6.5 Legal and Compliance

MediBook shall be configured to protect patient personal and medical information according to applicable hospital policies and legal requirements. The exact data-protection and legal compliance regime applicable to the deployment shall be confirmed by hospital management/legal counsel under TBD-10. Payment-processing compliance responsibilities shall be confirmed with the selected payment provider under TBD-05.

6.6 Auditability

The system shall retain sufficient operational information to determine when relevant appointment, payment, doctor-profile, and diagnosis-note changes occurred and which authorized account initiated them. The final audit-event list and retention period are tracked as TBD-11.

7 Verification

Each requirement shall be verified using one or more of the following methods:

- **Test:** executed against running code through unit, integration, system, or acceptance tests.
- **Analysis:** verified through calculation, modeling, configuration review, or technical analysis when direct testing is not practical.
- **Inspection:** verified by visual or manual examination of code, configuration, UI, or documentation.
- **Demonstration:** verified by observing the implemented system perform the required behavior during a sprint review or stakeholder demonstration.
- **Status:** Not Started / In Progress / Passed / Failed.

Req/Story ID	Verification Method	Test/Artifact Link	Status	Sprint/Release
US-1.1	Test	TC-REG-001	Not Started	Sprint 1
US-1.2	Test	TC-AUTH-001	Not Started	Sprint 1
US-1.3	Test / Inspection	TC-SEC-001	Not Started	Sprint 1
US-2.1 / US-2.2	Test	TC-SEARCH-001	Not Started	Sprint 1
US-2.3	Test	TC-AVAIL-001	Not Started	Sprint 1
US-3.1	Test	TC-BOOK-001	Not Started	Sprint 2
US-3.2	Test / Analysis	TC-SLOT-001	Not Started	Sprint 2
US-3.3	Test	TC-NOTE-001	Not Started	Sprint 2
US-4.1	Test	TC-PAY-001 (gateway sandbox)	Not Started	Sprint 2
US-4.2	Test	TC-PAY-002	Not Started	Sprint 2
US-4.3	Test	TC-PAY-003	Not Started	Sprint 2
US-5.1 / US-5.2	Test / Demonstration	TC-DR-001	Not Started	Sprint 3
US-5.3	Test / Inspection	TC-DIAG-001	Not Started	Sprint 3
US-6.1	Test	TC-REC-001	Not Started	Sprint 3
US-6.2	Test / Demonstration	TC-REC-002	Not Started	Sprint 3
US-7.1 / US-7.2	Test	TC-ADMIN-001	Not Started	Sprint 1
US-8.1 / US-8.2	Test	TC-EMAIL-001	Not Started	Sprint 2
US-9.1	Test / Demonstration	TC-HISTORY-001	Not Started	Sprint 4
US-9.2	Test / Inspection	TC-HISTORY-002	Not Started	Sprint 4
NFR-PERF-01 / NFR-PERF-02	Analysis / Test	PERF-001	Not Started	Release 1.0
NFR-SEC-01 / NFR-SEC-02	Inspection / Test	SEC-001	Not Started	Release 1.0
NFR-AVAIL-01	Analysis	SLA-001	Not Started	Release 1.0
BR-08 / BR-11	Test / Analysis	TC-CONC-001 / TC-PAY-003	Not Started	Sprint 2

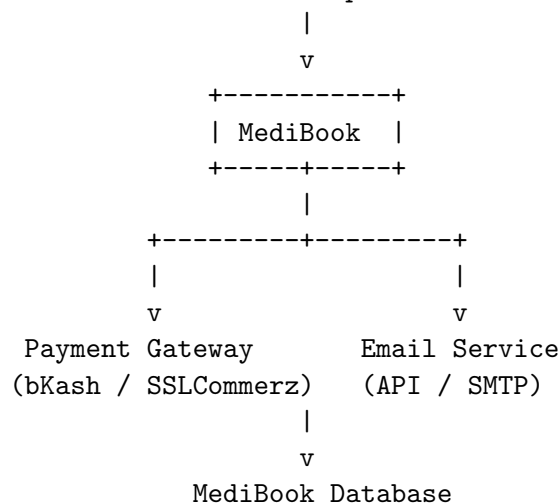
Appendix A: Glossary

Term	Definition
MediBook	The web-based appointment booking and management system developed for City General Hospital.
Appointment	A scheduled consultation between a patient and doctor at a specified date and time.
Slot	A specific date/time window during which a doctor is available for one appointment.
Pending Payment	Booking status indicating a slot has been reserved but payment has not yet been successfully completed.
Confirmed	Booking status indicating payment has succeeded and the appointment is secured.
Released	Booking status indicating the pending appointment was not completed successfully and the slot is available again.
Payment Gateway	External service such as bKash or SSLCommerz that processes online payments and returns the payment result.
Email Notification Service	External transactional email API or SMTP service used to send booking confirmations and appointment reminders.
BDT / Taka	Bangladeshi Taka, the currency in which consultation fees and payments are denominated.
BST	Bangladesh Standard Time, UTC+6.
PCI-DSS	Payment Card Industry Data Security Standard, a security standard relevant to payment-card processing.
Diagnosis Notes	Clinical notes entered by an authorized doctor describing the outcome of a patient consultation.
MoSCoW	Prioritization method using Must, Should, Could, and Won't categories.

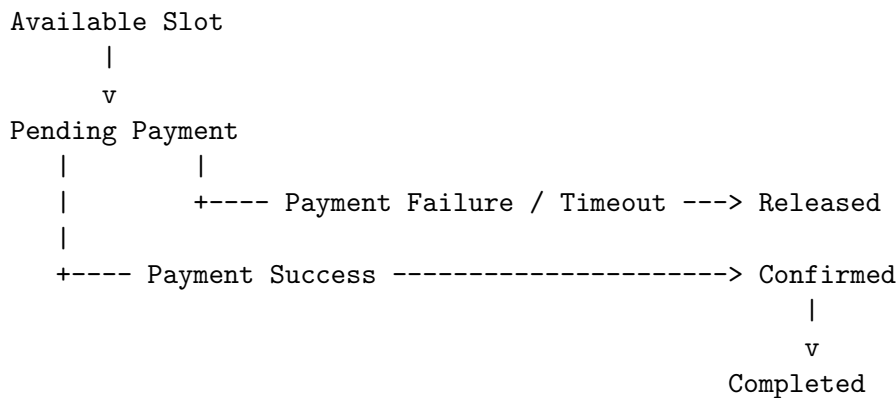
Appendix B: Analysis Models

B.1 System Context

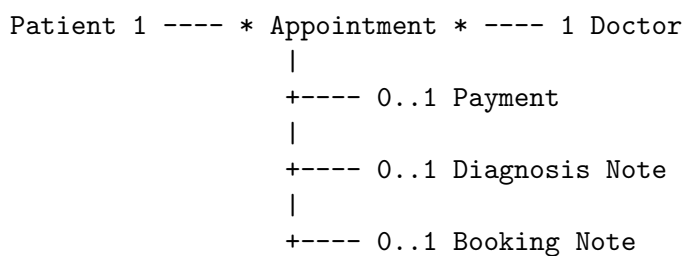
Patient / Doctor / Receptionist / Administrator



B.2 Appointment State Model



B.3 High-Level Data Model



Appendix C: To Be Determined List

1. TBD-01: Final production Payment Gateway selection (bKash or SSLCommerz) and provider-specific API/callback configuration.
2. TBD-02: Final transactional Email Notification Service and its API/SMTP configuration.
3. TBD-03: Maximum Pending Payment duration before an unpaid slot is automatically released.
4. TBD-04: Session inactivity timeout duration and related session-security settings.
5. TBD-05: Exact payment-security/compliance boundary and responsibilities between MediBook and the selected payment gateway, including PCI-DSS applicability.
6. TBD-06: Formal production uptime target/SLA; the proposed target is 99.5%.
7. TBD-07: Production hosting provider, server operating system, database technology, backup, and disaster-recovery configuration.
8. TBD-08: Whether Bangla-language support is required for Version 1.0 or a later release.
9. TBD-09: Required data-retention period for patient, appointment, payment-reference, and diagnosis information.
10. TBD-10: Applicable data-protection and legal compliance requirements for the hospital deployment.
11. TBD-11: Required audit events and audit-log retention period.
12. TBD-12: Appointment reminder interval, such as 24 hours before the scheduled appointment.
13. TBD-13: Final user-documentation delivery format and whether PDF copies are required.